Adei *et al. Discover Public Health* (2026) 23:153
https://doi.org/10.1186/s12982-026-01410-6

Discover Public Health

RESEARCH Open Access

# Association between domains of social support and healthcare utilisation among informal caregivers of older adults in slums, Ghana

Dina Adei[1*], Williams Agyemang-Duah[2], Bright Osei Boateng[1] and Anthony Acquah Mensah[3]

*Correspondence:
Dina Adei
ladydeeust@yahoo.com
[1]Department of Planning, Kwame Nkrumah of Science and Technology, Kumasi, Ghana
[2]Department of Public Health Sciences, Queen's University, Kingston, ON K7L 3N6, Canada
[3]Department of Sociology, Trent University, Peterborough, ON K9L 0G2, Canada

## Abstract

**Background** Informal caregivers play a crucial role in supporting older adults; however, their health needs remain overlooked, potentially hindering progress toward health-related Sustainable Development Goals, particularly Goal 3. As ageing populations and disease burdens increase, prioritising the well-being of informal caregivers in resource-poor settings is important. This study aims to understand how the social environment, particularly social support, influences healthcare utilisation among informal caregivers of older adults living in slum areas in Ghana.

**Methods** The study recruited a sample of 458 informal caregivers of older adults living in slum communities within the Greater Kumasi metropolis of Ghana. Employing the binary logistic regression models, the association between domains of social support and healthcare utilisation was estimated. The analysis involved utilising odds ratio and confidence interval , with statistical significance set at 0.05 or lower.

**Results** The study revealed that financial support (OR: 1.914, 95% CI 1.249–2.934) had a positive and statistically significant association with healthcare utilisation. In contrast, emotional support (OR: 1.147, 95% CI 0.707-1.862), instrumental support (OR: 1.368, 95% CI 0.827- 2.264), and informational support (OR: 0.704, 95% CI 0.432-1.148) showed no significant association with healthcare utilisation. The study showed that those with basic education (OR: 4.243, 95% CI 1.259–14.298), those who have never been ill (OR: 4.566, 95% CI 2.715–7.678), and those not diagnosed with NCDs were significantly more likely to utilise healthcare than their counterparts.

**Conclusion** The results substantially enhance our understanding of the association between financial support and healthcare utilisation among informal caregivers in slum areas. Furthermore, the findings inform the design of targeted healthcare interventions that specifically address the distinctive financial and socioeconomic challenges faced by informal caregivers, thereby contributing to more effective and responsive healthcare strategies in these vulnerable communities.

**Keywords** Social support, Healthcare utilization, Informal caregivers, Older adults, Slum communities, Ghana

## 1 Introduction

Informal caregiving is central to supporting the well-being of older adults [1]. In Ghana, informal caregiving is predominantly family-based, with an estimated 70% of older adults receiving care from relatives, primarily due to the limited availability of formal elderly care services [2]. Informal care encompasses unpaid support from family, friends, or volunteers who assist individuals with chronic illnesses or disabilities [3]. The increasing population of older adults in Ghana has implications for informal caregivers, as older individuals with health concerns require prolonged care [4]. In this study, informal caregivers are defined as unpaid individuals, including family members, friends, or community members, who support older adults aged 60 years and older, by assisting with daily activities, providing healthcare, and social support [5]. This form of care is widespread and serves as a critical component of the healthcare system in Ghana by reducing costs and improving the quality of life for care recipients [6].

Social support impacts caregiver well-being [7]. According to Klein et al. [8], social support is the component of interpersonal connections that makes the recipient feel appreciated by others. In this study, social support is considered across four domains: emotional, instrumental, informational, and financial, which affect whether and how caregivers access formal healthcare. Emotional support enhances well-being [9], while instrumental support facilitates access to healthcare, especially in economically disadvantaged areas [10]. Informational support helps caregivers make informed healthcare decisions and self-efficacy [11]. Financial support directly and indirectly reduces the burden of healthcare costs associated with seeking healthcare [2]. These domains of social support are conceptually distinct and may not uniformly influence health utilisation among caregivers. Healthcare utilisation refers to the extent to which caregivers access healthcare services, such as medical check-ups, preventive care, and treatment for illnesses, for their own well-being [12]. Access to healthcare is vital for caregivers, as their ability to provide care is directly linked to their own health status [13]. At the same time, some social supports can facilitate timely healthcare use, whereas others may unintentionally delay access to formal healthcare [14].

Slum environments present unique barriers and pressures that can influence the effect of social support on healthcare utilisation [15]. Slums in Ghana are characterised by overcrowded housing, land tenure insecurity, and inadequate access to basic services such as potable water, sanitation, drainage, and waste management [5, 16]. Informal caregivers in slums face barriers such as financial constraints, inadequate healthcare infrastructure, social isolation, and precarious living conditions, which can shape their social support networks and healthcare utilisation patterns [17]. While studies have revealed the role of social support in shaping healthcare utilisation and psychological well-being among older adults, the quality and relevance of support may differ [18, 19]. Also, in resource-poor environments, such as slums in Ghana, social support may operate differently than in non-slum urban and rural areas.

In Ghana, previous studies on informal caregivers of older adults have examined healthcare utilisation using the health belief model [20], the relationship between caregiver burden and health-related quality of life [21], geographical location and caregiver burden [22], and gender and health insurance enrollment [23]. Also, studies have either focused on general populations and urban areas [8, 24, 25]. Nevertheless, evidence that links the distinct domains of social support to caregivers' healthcare utilisation in slum communities is scarce. The foregoing limits policy interventions that aim to strengthen informal caregivers' well-being in resource-poor urban environments in Ghana. This study, therefore, addresses this gap by focusing on informal caregivers of older adults in slum communities within the Greater Kumasi Metropolitan Area (GKMA), Ghana.

This study aims to achieve two objectives: (i) to assess the prevalence of healthcare utilisation among informal caregivers of older adults residing in slums in GKMA, and (ii) to examine the relationship between the domains of social support and healthcare utilisation. By examining the relationship among the domains of support and employing a count-based measure of utilisation, the study provides details of which forms of support are associated with informal caregivers' healthcare utilisation. The contribution of the study is practical and policy-oriented. First, it clarifies the roles of emotional, instrumental, informational, and financial support that shape caregivers' formal healthcare use in slum areas. Second, the findings inform community-level strategies that leverage social support to facilitate access to healthcare for caregivers and provide evidence to support responsive health policies and services in slum communities.

## 2 Methods

### 2.1 Study design and settings

This study is based on primary data collected through an extensive cross-sectional survey designed to evaluate the healthcare needs of informal caregivers (aged 18 years and older) of older adults aged 60 years and above in Ghana. The survey, conducted between June and August 2023 in the Greater Kumasi Metropolitan Area (GKMA), gathered firsthand data directly from informal caregivers in GKMA. Eight communities within the metropolitan area were selected using simple random sampling techniques to ensure representativeness. The study's inclusion criteria were: (i) informal caregivers aged 18 years and older providing care for older adults aged 60 years and above in selected communities within the GKMA for at least six months, and (ii) informal caregivers who were mentally and physically capable of responding to the survey questions and providing informed consent. The study excluded professionals or formal caregivers, caregivers of persons less than 60 years, and those who have been providing care for less than six months. Also, informal caregivers residing outside the selected communities or those unable to communicate effectively due to severe cognitive impairment or language barriers were excluded.

### 2.2 Sampling and strategy

A multi-stage sampling approach was utilised to identify participants. First, two municipalities in the Ashanti Region, Oforikrom Municipality and Asokore Mampong Municipality, were purposively chosen based on the following criteria: (i) the population of older adults aged 60 years and above (ii) the presence of slums or informal settlements, and (iii) the availability of population data for various age groups of older adults within the municipalities. Next, eight communities were randomly selected, comprising four urban and four rural areas. These eight communities accounted for a third of the communities in the two municipalities for which the Ghana Statistical Service has provided census data [26]. The selection of four urban and four rural communities aimed to ensure that the study sample was representative of the broader population. Additionally, two slum communities, Aboabo (in the Asokore Mampong Municipality) and Ayigya (in the Oforikrom Municipality), were chosen to align with the focus of the study. At the participant level, snowball sampling was used in each slum community to identify caregivers of older adults. The use of snowballing was due to the unavailability of data on the number of informal caregivers in the study area and Ghana [20]. The snowball sampling technique was applied by initially identifying informal caregivers who met the inclusion criteria and community members who knew of caregivers and inviting them to refer other eligible caregivers. This approach was necessary due to the absence of a reliable database on informal caregivers in the study area. The multi-stage sampling approach, a combination of probability and non-probability sampling approaches, allowed for the selection of communities and a pragmatic approach to recruit informal caregivers.

To determine the minimum sample size, we applied the statistical formula recommended by Lwanga and Lemenshow [27]: n = design effect × [(Zα/2)2 ×P (1-P)]/ε2. Considering a design effect of 1.5, a 95% confidence interval, and a 4% margin of error, the calculation yielded a minimum sample size of 899. Given the uncertainty surrounding the exact number of informal caregivers in the study area and Ghana, a conservative prevalence (p) of 0.48 was adopted, following the approach outlined by Agyemang-Duah & Rosenberg [20], resulting in the determination of 899 as the minimum sample size for informal caregivers of older adults in slum areas, as indicated by Adei et al. [5]. Considering a 5% non-response rate, we initially expected a final sample size of 944 participants. However, 70 individuals (7.42%) declined to participate, 26 (2.75%) provided incomplete responses, and 9 (0.95%) had missing data, resulting in a response rate of 88.88% (839 participants). A total of 458 participants were sampled from slum communities selected for this study. The distribution of informal caregivers sampled from the slum communities was 200 and 258 from Aboabo and Ayigya, respectively.

### 2.3 Data collection instrument and procedures

A structured questionnaire was used in this study. The questionnaire encompassed inquiries related to demographics, socio-economic factors, health characteristics, and food security status, among others. To ensure the quality and reliability of the measurement tools, a pilot study was conducted with 20 informal caregivers from Ayeduase and Kotei communities, which share similar characteristics with the study areas. The feedback from the pilot study enhanced clarity, consistency, and relevance of the questionnaire items. This study used an interviewer-administered survey questionnaire. Research assistants underwent a two-week training session to ensure consistency in translations and question delivery on the questionnaire. Additionally, field supervisors monitored data collection to strengthen adherence to the standardised approach. Research assistants conducted face-to-face interviews to ensure that participants with limited or no formal education could understand and respond to the questions. The questionnaire was translated into Twi, a widely spoken local language in Ghana, and the GKMA. The administration of the questionnaire lasted approximately 30–45 minutes per participant. Responses were inputted using the KoboToolbox platform to ensure real-time data entry and reduce errors during data collection. Data collection was conducted in a manner that prioritised participant privacy, with interviews taking place in a quiet and private setting within the participant's residence. This was particularly important for informal caregivers who lived with their care recipients to minimise interruptions and ensure privacy.

### 2.4 Measurement of variables

Healthcare utilisation, the outcome variable in this study, was defined as seeking treatment from a formal healthcare provider (specifically clinics, hospitals, and health centres) within the past six months. Healthcare utilisation was measured as a count variable, that is, the self-reported number of times that a participant sought treatment for an illness in the past six months. The measurement of healthcare utilisation as a count variable is consistent with Gyasi et al. [18] and Agyemang-Duah [20]. These responses were coded as: 0 = never, 1 = once, 2 = two times, 3 = three times, 4 = four times, and 5 = five or more times. However, because of over-dispersion in the fitted models, we recategorised the responses as yes/no: those who responded never were coded as No, and those who responded once, two times, three times, four times, and five or more times were coded as Yes. This is consistent with the literature on healthcare utilisation, which has measured healthcare utilisation as a dichotomous variable [28, 29]. Illness was defined as both acute and chronic health conditions, including malaria and other infectious diseases.

The primary predictor variable was social support. Social support in this study was assessed across four domains: emotional, instrumental, informational, and financial support. Emotional support refers to the availability of individuals who provide comfort, encouragement, and reassurance to caregivers [30]. Instrumental support involved practical assistance, such as help with caregiving duties, transportation, or household tasks [31]. Informational support encompassed access to advice, guidance, or knowledge related to caregiving responsibilities and healthcare services. Financial support refers to the provision of money in the form of monetary or material assistance to ease the economic burden of caregiving [32]. An adapted version of the Multidimensional Scale of Perceived Social Support was used to measure social support in this study [33]. The original 12-item scale was modified to reflect Ghana's local and cultural context. Responses were recorded on a 5-point Likert scale: 0 for 'never', 1 for 'less frequently', 2 for 'frequently', 3 for 'very frequently', and 4 for 'every day'. The Cronbach's alpha value was 0.838.

Beyond the domains of social support, demographic, socio-economic, and health-related variables were controlled for. Demographic variables were gender (coded as 0 for male and 1 for female), age (in years) (coded as 0 for 18–29, 1 for 30–39, 2 for 40–49, 3 for 50–59, 4 for 60 or above), ethnicity (coded as 0 for Akan, 1 for non-Akan), religion (coded as 0 for Christian, 1 for non-Christian), and marital status (coded as 0 for married, 1 for not married). Socio-economic variables included education (coded as 0 for no formal education, 1 for basic education, 2 for secondary education, 3 for tertiary education), compensation for informal care (coded as 0 for no, 1 for yes), and enrolment in a national health insurance scheme (coded as 0 for no, 1 for yes) for informal caregivers. Health-related variables comprised the frequency of illness (coded as 0 for never, 1 for frequent) and the presence of chronic non-communicable diseases (coded as 0 for no, 1 for yes). In this study, chronic non-communicable diseases (NCDs) encompass conditions such as hypertension, diabetes, and cardiovascular diseases. The literature on healthcare utilisation informed the selection of these control variables [13]. The presence of multicollinearity in the control and independent variables was assessed using the Variance Inflation Factor (VIF). The VIF values were less than 3.5 for all variables, indicating no significant multicollinearity (see Table 2).

### 2.5 Data analysis

Utilising Statistical Package for the Social Sciences (SPSS) software version 25, our data analysis employed a combination of descriptive and inferential statistical techniques. Descriptive methods, such as percentages and frequencies, were applied to provide insights into the demographic, socio-economic and health characteristics of the sample participants. For inferential analysis, binary logistic regression models were utilised to examine the relationships between the outcome and independent variables. Two separate models were established: Model 1 examined the crude association between social support domains and healthcare utilisation, while Model 2 adjusted for demographic, socio-economic, and health-related variables to enhance model robustness. Odds Ratio (OR) and Confidence Interval (CI) were reported. Statistical significance was set at a probability value of 0.05 or lower.

### 2.6 Ethical consideration

Ethical approval was obtained from the Committee on Human Research, Publication, and Ethics at the Kwame Nkrumah University of Science and Technology, Kumasi, Ghana (Ref. CHRPE/AP/528/23). Participants were informed about the study's objectives, and their verbal or written consent was sought. Verbal and written consent were documented with a signature or thumbprint. Participants were guaranteed privacy and anonymity, with their involvement being entirely voluntary. Participants were also advised of their right to abstain from answering any questions, and they had the option to withdraw from the study at any time without facing any penalties. Participants were not compensated for their participation in the study.

## 3 Results

### 3.1 Sample characteristics of the participants

Table 1 presents information on the sample characteristics of the respondents. The results showed that 88.4% of the participants were female, 37.3% were aged between 40 and 49 years, 72.7% were of Akan ethnicity, 75.1% were Christian, and 81.4% were married. The data further indicated that 45.4% of the respondents had basic education, 96.3% did not receive pay for the care they provided, 72.1% had enrolled in the national health insurance scheme, 53.1% frequently got ill, and 20.1% had been diagnosed with chronic non-communicable diseases (see Table 1).

### 3.2 Prevalence of healthcare utilisation

Figure 1 presents the prevalence of healthcare utilisation among informal caregivers of older adults. Approximately 40.4% of the participants had never sought healthcare in the past six months, 15.5% sought healthcare once in the last six months preceding the survey, and 14.8% sought healthcare 3 times in the last six months before the study.

### 3.3 Association between the domains of social support and healthcare utilisation

Table 2 presents information on the association between social support domains and healthcare utilisation. In Model 1, the results showed financial support (OR:2.598, 95% CI 1.808–3.732) was positively and statistically significantly associated with healthcare utilisation. However, other social support domains, such as emotional support (OR: 1.154, 95% CI 0.747- 1.784), instrumental support (OR: 1.123, 95% CI 0.730 – 1.730), and

**Table 1** Sample characteristics of the participants

| Variables | Category | Frequency | % |
|---|---|---|---|
| Gender | Male | 53 | 11.6 |
| | Female | 405 | 88.4 |
| Age (years) | 18–29 | 61 | 13.3 |
| | 30–39 | 96 | 21.0 |
| | 40–49 | 171 | 37.3 |
| | 50–59 | 104 | 22.7 |
| | 60 or above | 26 | 5.7 |
| Ethnic group | Akan | 333 | 72.7 |
| | Non-Akan | 125 | 27.3 |
| Religion | Christian | 344 | 75.1 |
| | Non-Christian | 114 | 24.9 |
| Marital status | Married | 373 | 81.4 |
| | Single | 85 | 18.6 |
| Education | No formal education | 128 | 27.9 |
| | Basic education | 208 | 45.4 |
| | Secondary education | 99 | 21.6 |
| | Tertiary education | 23 | 5.0 |
| Compensated for care provided | No | 441 | 96.3 |
| | Yes | 17 | 3.7 |
| Registered on NHIS | No | 128 | 27.9 |
| | Yes | 330 | 72.1 |
| Frequency of illness | Never | 215 | 46.9 |
| | Frequent | 243 | 53.1 |
| Diagnosed with any chronic NCDs | No | 366 | 79.9 |
| | Yes | 92 | 20.1 |
| | Total | 458 | 100.0 |

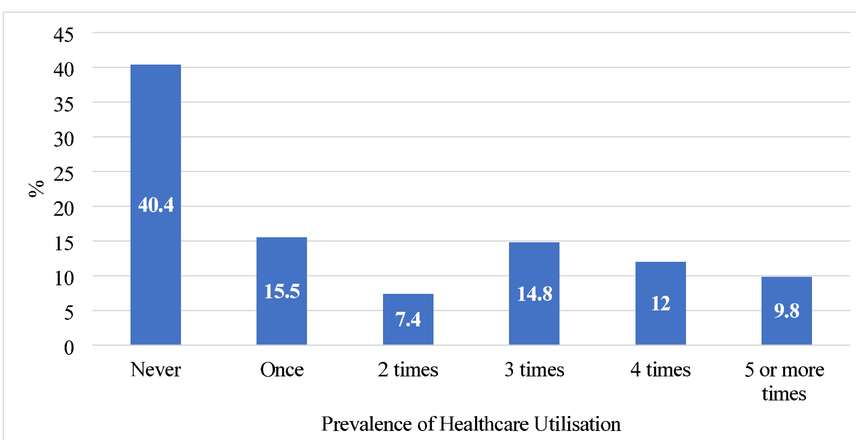


**Fig. 1** Prevalence of healthcare utilisation

informational support (OR: 0.802, 95% CI 0.527- 1.221), were not statistically significantly associated with healthcare utilisation.

In Model 2 (final model), the results revealed that financial support (OR: 1.914, 95% CI 1.249–2.934) was positively and statistically significantly associated with healthcare utilisation, although emotional support (OR: 1.147, 95% CI 0.707-1.862), instrumental support (OR: 1.368, 95% CI 0.827- 2.264) and informational support (OR: 0.704, 95% CI 0.432-1.148) were not statistically significant. Beyond financial support, socioeconomic

**Table 2** Association between domains of social support and healthcare utilisation

| | **Model 1** | | | **Model 2 (final 2)** | | | **Collinearity statistics** | |
|---|---|---|---|---|---|---|---|---|
| | **95% wald confidence interval for exp (B)** | | | **95% wald confidence interval for exp (B)** | | | | |
| | **OR** | **Lower** | **Upper** | **OR** | **Lower** | **Upper** | **Tolerance** | **VIF** |
| Emotional support | 1.154 | 0.747 | 1.784 | 1.147 | 0.707 | 1.862 | 0.522 | 1.915 |
| Instrumental support | 1.123 | 0.730 | 1.730 | 1.368 | 0.827 | 2.264 | 0.506 | 1.977 |
| Informational Support | 0.802 | 0.527 | 1.221 | 0.704 | 0.432 | 1.148 | 0.490 | 2.041 |
| Financial support | **2.598*** | **1.808** | **3.732** | **1.914*** | **1.249** | **2.934** | 0.488 | 2.048 |
| Gender | | | | | | | 0.905 | 1.105 |
| Male | | | | 0.746 | 0.344 | 1.615 | | |
| Female (ref) | | | | 1.00 | | | | |
| Age (years) | | | | | | | 0.508 | 1.967 |
| 18–29 | | | | 1.068 | 0.247 | 4.612 | | |
| 30–39 | | | | 0.316 | 0.094 | 1.064 | | |
| 40–49 | | | | 1.044 | 0.367 | 2.968 | | |
| 50–59 | | | | 1.201 | 0.413 | 3.492 | | |
| 60 or above (ref) | | | | 1.00 | | | | |
| Ethnicity | | | | | | | 0.313 | 3.200 |
| Akan | | | | 0.932 | 0.371 | 2.340 | | |
| Non-Akan (ref) | | | | 1.00 | | | | |
| Religion | | | | | | | 0.312 | 3.203 |
| Christian | | | | 2.411 | 0.912 | 6.374 | | |
| Non-Christian (ref) | | | | 1.00 | | | | |
| Marital status | | | | | | | 0.560 | 1.785 |
| Married | | | | 1.238 | 0.483 | 3.176 | | |
| Single (ref) | | | | 1.00 | | | | |
| Education | | | | | | | 0.618 | 1.617 |
| No formal education | | | | 2.379 | 0.660 | 8.580 | | |
| Basic education | | | | **4.243*** | **1.259** | **14.298** | | |
| Secondary education | | | | 3.047 | 0.931 | 9.974 | | |
| Tertiary education (ref) | | | | 1.00 | | | | |
| Compensated for care provided | | | | | | | 0.921 | 1.086 |
| No | | | | 3.704 | 0.703 | 19.526 | | |
| Yes (ref) | | | | 1.00 | | | | |
| NHIS enrollment | | | | | | | 0.871 | 1.148 |
| No | | | | 1.018 | 0.933 | 1.111 | | |
| Yes (ref) | | | | 1.00 | | | | |
| Frequency of illness | | | | | | | | |
| Never | | | | **4.566*** | **2.715** | **7.678** | 0.674 | 1.483 |
| Frequent (ref) | | | | 1.00 | | | | |
| Experience with chronic NCDs | | | | | | | 0.788 | 1.269 |
| No | | | | **2.769*** | **1.362** | **5.627** | | |
| Yes (ref) | | | | 1.00 | | | | |
| (Scale) | | | | | | | | |
| Model fitting information | | | | | | | | |
| Likelihood Ratio Chi-Square (Omnibus Test) (sig.) | 49.784 (0.000) | | | | | 165.290 (0.000) | | |
| Wald Chi-square (sig) | 34.379 (0.000) | | | | | 26.662 (0.000) | | |

**Table 2** (continued)

| | Model 1 | | | Model 2 (final 2) | | | Collinearity statistics | |
|---|---|---|---|---|---|---|---|---|
| | 95% wald confidence interval for exp (B) | | | 95% wald confidence interval for exp (B) | | | | |
| | OR | Lower | Upper | OR | Lower | Upper | Tolerance | VIF |
| Pearson chi-square value/df | 2.364 | | | | | 1.082 | | |
| Cox & Snell R square | 0.221 | | | | | 0.303 | | |
| Nagelkerke R square | 0.298 | | | | | 0.409 | | |
| 2 Log likelihood | -503.529 | | | | | -452.619 | | |
| Hosmer and Lemeshow Chi-square (sig) | 1.032 (0.905) | | | | | 0.16 | | |

*Statistical significance was observed at the 0.05 level

and health-related factors were associated with healthcare utilisation among participants. First, the study found that those with basic education were statistically significantly 4.243 times more likely to utilise healthcare services compared to those with a tertiary education (OR: 4.243, 95% CI 1.259–14.298). Second, the findings of the study suggested that those who had never been ill were 4.566 times statistically significantly less likely to utilise healthcare compared to those who frequently get ill (OR: 4.566, 95%CI 2.715–7.678). Lastly, those who had not been diagnosed with NCDs were 2.769 times significantly more likely to utilise healthcare compared to those who had been diagnosed with NCDs (OR: 2.769, 95%CI 1.362–5.627) (see Table 2).

## 4 Discussion

To our knowledge, this study extends beyond conventional research on social support by examining a wide range of factors that influence both social support dynamics and healthcare utilisation patterns among informal caregivers of older adults. In the final model (2), the study found that financial support was positively and statistically significantly associated with healthcare utilisation, underscoring its role in influencing informal caregiver's utilisation of healthcare services. The positive association between financial support and healthcare utilisation highlights the well-established link between financial resources and healthcare access. Financial barriers, including the cost of medical care, transportation, and medications, are recognised determinants of healthcare utilisation [34, 35]. For instance, in the Ghanaian context, earlier studies have shown that individuals who report financial barriers to healthcare are more likely to delay seeking care, emphasising the importance of adequate financial support for regular healthcare use [13, 20]. This is because adequate financial support can alleviate these barriers, enabling individuals to seek timely and necessary healthcare services. Consequently, individuals with financial support have higher healthcare utilisation, as they can cover transportation, treatment, and medication costs associated with healthcare utilisation. This evidence suggests that strengthening the financial support for informal caregivers is likely to improve their healthcare utilisation [13].

Beyond the financial domain of social support, our findings further indicate that individuals with lower levels of education, specifically those with basic education, were more likely to utilise formal healthcare services compared to those with tertiary education, suggesting a relationship between educational attainment and healthcare utilisation among informal caregivers of older adults in slum communities. Higher health literacy is associated with improved adherence to medical advice, preventive health behaviours, and timely healthcare-seeking behaviour [36]. The result indicates strong financial support and resourcefulness within the community, in which informal caregivers, regardless of education level, have access to health-related knowledge on health promotion and disease prevention, thereby enhancing their health literacy and healthcare utilisation [37, 38]. Informal networks provide a nurturing social support system that supports an environment conducive to encouraging health-promoting behaviours.

This study revealed that NCD experiences and frequency of illness were significantly associated with healthcare utilisation among participants. More specifically, findings from this study have shown that the participants who had not been diagnosed with NCDs and those who had never been ill were significantly more likely to utilise healthcare services compared to those who were diagnosed with NCDs and those who had never been ill. This finding is relatively new in the caregiving-healthcare literature, particularly in the Ghanaian literature, and underscores a contribution to the field. The mechanisms by which individuals without experience of NCDs or illness report higher healthcare utilisation are as follows. First, differences in how individuals value their health might explain this finding. People without chronic diseases may perceive that regular treatment reduces their risk of conditions like hypertension, cancer, or diabetes. In contrast, those with a diagnosed chronic illness might not share this view. Such individuals believe ongoing healthcare helps them stay healthy and prevents chronic diseases, potentially increasing healthcare demand among them. Lastly, barriers such as financial and transportation issues can limit healthcare use even in the presence of diseases, as widely discussed in healthcare research [39, 40] and could strongly reinforce this finding. Even when diagnosed with chronic diseases, individuals facing these barriers may be less likely to seek care, as the health belief model indicates that such barriers reduce healthcare-seeking behaviour [41, 42]. These points suggest that individuals without a diagnosis of chronic conditions are likely to have better socioeconomic access to financial and transportation resources, which supports their healthcare use, reduces barriers, and increases their engagement with healthcare services. Therefore, strengthening the socio-economic positions of informal caregivers, particularly those who are diagnosed with chronic diseases, could improve their healthcare utilisation.

## 5 Policy and practice implications

The identification of the financial support, education, and health-related factors associated with healthcare utilisation among participants emphasise the need for policy and practice implications to address financial, socio-economic, and health disparities and to improve overall healthcare access among informal caregivers of older adults residing in slum communities in Ghana. Policies should prioritise strategies to strengthen the financial capacities of informal caregivers of older adults. Thus, reducing financial barriers to healthcare among participants remains paramount to scaling up healthcare utilisation among informal caregivers of older adults in Ghana.

Informal caregivers play a crucial role in the healthcare system; therefore, policymakers should prioritise the financial support system and well-being of informal caregivers, considering the implementation of support programs, respite care initiatives, and educational resources. Integrating caregiver-specific considerations into healthcare policies can help create a supportive environment for this essential group. Education emerges as a key determinant of healthcare utilisation, with individuals having basic education, being more likely to utilise healthcare services compared to those with tertiary education. Policies should focus on bridging educational disparities by promoting health literacy across all educational levels. Public health campaigns should emphasise preventive care and routine health check-ups, targeting individuals with higher educational attainment to ensure a comprehensive approach to healthcare promotion.

Also, healthcare providers should integrate assessments of patients' financial support systems into care plans, recognising the importance of addressing not only the presence but also the quality and context of support. Policies should encourage healthcare providers to engage in patient-centred care that includes discussions about health perceptions. Public health initiatives should emphasise the importance of preventive healthcare measures, even for individuals who perceive themselves as generally healthy.

Policies should emphasise equitable access to healthcare services, ensuring that socioeconomic status does not impede healthcare utilisation. Policymakers can contribute to a more inclusive healthcare system by addressing financial barriers and ensuring access for all, regardless of economic circumstances. Continuous monitoring and evaluation of healthcare utilisation patterns are essential components of effective policy implementation. Regular data collection and analysis will enable policymakers to assess the impact of interventions, identify evolving trends, and make informed adjustments to healthcare policies. This iterative process ensures that policies remain responsive to the dynamic nature of healthcare utilisation within the studied population.

### 5.1 Limitations of the study

Despite the insights gained from this study, certain limitations should be acknowledged. First, the use of snowball sampling introduces potential selection bias as participants referred by others may share similar characteristics. The sample in this study, therefore, may not reveal the diversity of informal caregivers in the study area. The findings should be interpreted with caution about generalisability. The study did not incorporate specific variables measuring slum-dwelling conditions, such as inadequate housing, poor sanitation, and overcrowding, which could impact healthcare utilisation patterns. This makes it difficult to differentiate the unique challenges faced by informal caregivers in slums compared to those in other urban areas. Second, marital status was broadly categorised into "married" and "single," grouping never-married, widowed, and divorced individuals, potentially overlooking differences in how these subgroups experience social and economic challenges that influence healthcare utilisation. Lastly, healthcare utilisation was assessed without distinguishing between different types of healthcare services, and the severity of health conditions was not accounted for, limiting the study's ability to provide a more nuanced understanding of caregivers' healthcare needs.

## 6 Conclusion

The findings of this study shed light on the multifaceted nature of healthcare utilisation, revealing significant associations with education, frequency of illness, and experiences of chronic NCDs, beyond the financial domain of social support. The identified patterns underscore the need for a holistic understanding of individuals' healthcare-seeking behaviours and the importance of tailoring interventions to address diverse determinants. Policies should prioritise the financial support and well-being of this group, recognising the potential impact of caregiving responsibilities on caregivers' health and healthcare utilisation patterns. Integrating caregiver-specific considerations into healthcare policies can contribute to a more supportive environment for this essential segment of the population. Furthermore, the study emphasise the need for policies that address educational disparities in healthcare utilisation. Policies can be designed to integrate assessments of patients' financial support systems into care plans, fostering a more personalised approach to healthcare delivery by recognising the influence of financial support dynamics on healthcare utilisation. Policies should prioritise equitable access to healthcare services, ensuring that socioeconomic status does not hinder healthcare utilisation. Future research endeavours should distinguish between different types of healthcare services utilised, such as preventive, emergency, or chronic disease management, to provide more insights.

## 7 Contributions to the literature

- Studies are yet to be conducted on the association between the domains of social support and healthcare utilisation among informal caregivers of older adults residing in slum communities in Ghana
- This study examines how the domains of social support influence healthcare utilisation among informal caregivers of older adults living in slum communities in Ghana
- This study highlights that the financial domain of social support significantly contributes to disproportional healthcare utilisation among the participants
- This study has the potential to contribute partly to the realisation of United Nations Sustainable Development Goal 3 (Good Health and Well-being)

**Acknowledgements**
We extend our deepest gratitude to all those who contributed to the successful completion of this research work. We also extend our acknowledgment to the authors and publishers whose works informed our research. We would like to express our heartfelt thanks to the participants of this research, whose willingness to share their experiences and insights has been crucial to the depth and richness of the findings. Additionally, we express gratitude to Miss. Pamela Konadu, Mr. Obeng Diawuo Van-ess, Mr. Opoku Mensah, Miss. Abigail Obiri Yeboah and Mr Anokye Pius Opoku for their valuable assistance in the data collection process. Your contributions have significantly enhanced the understanding of social support and healthcare utilisation dynamics among informal caregivers in slum communities.

**Author contributions**
Conceptualization, DA and WA-D; methodology, DA, WA-D and BOB; software, WA-D, and AAM; formal analysis, WA-D.; data curation, DA and BOB; writing—original draft preparation, DA; writing—review and editing, DA, WA-D, BOB and AAM; supervision, DA.; funding acquisition, DA. All authors have read and agreed to the published version of the manuscript.

**Funding**
This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

**Data availability**
The datasets utilised or analysed in this present study can be obtained from the corresponding author upon reasonable request.

## Declarations

**Ethics approval and consent to participate**
All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. This study strictly adhered to rigorous ethical standards, obtaining clearance from the Committee on Human Research, Publication, and Ethics at Kwame Nkrumah University of Science and Technology, Kumasi, Ghana (Ref. CHRPE/AP/528/23). To uphold transparency and participant respect, we communicated the study's objectives and secured both oral and written informed consent before initiating data collection. Participants were explicitly briefed on the voluntary nature of their involvement, assuring them the option to refrain from answering sensitive or personal questions and withdraw from the study at any point without facing penalties.

**Consent for publication**
Not applicable.

**Competing interests**
The authors declare no competing interests.

Received: 15 September 2025 / Accepted: 16 January 2026
Published online: 06 February 2026

**References**

1. MacKenzie A, Greenwood N. (2012). Positive experiences of caregiving in stroke: a systematic review. In Disability and Rehabilitation (Vol. 34, Issue 17, pp. 1413–1422). https://doi.org/10.3109/09638288.2011.650307
2. Agyemang-Duah W, Rosenberg MW. Geographical location as a determinant of caregiver burden: a rural-urban analysis of the informal caregiving, health, and healthcare survey in Ghana. BMC Prim Care. 2024;25:317. https://doi.org/10.1186/s12875-024-02562-7.
3. Agree EM, Glaser K. (2009). Demography of Informal Caregiving. In International Handbook of Population Aging (pp. 647–668). Springer Netherlands. https://doi.org/10.1007/978-1-4020-8356-3_29
4. Roth DL, Fredman L, Haley WE. Informal caregiving and its impact on health: a reappraisal from population-based studies. Gerontologist. 2015;55(2):309–19. https://doi.org/10.1093/geront/gnu177.
5. Adei D, Agyemang-Duah W, Boateng BO, et al. Predictors of food security status among informal caregivers of older adults residing in slums in Ghana. BMC Public Health. 2025;25:447. https://doi.org/10.1186/s12889-025-21666-y.
6. Trujillo AJ, Mroz TA, Piras C, Angeles G, Tran N. Caregiving and elderly health in Mexico. Int J Health Serv. 2012;42(4):667–94. https://doi.org/10.2190/HS.42.4.f.
7. Gyasi RM, Phillips DR, Abass K. Social support networks and psychological wellbeing in community-dwelling older Ghanaian cohorts. Int Psychogeriatr. 2019;31(7):1047–57. https://doi.org/10.1017/S1041610218001539.
8. Klein TM, Andrees V, Kirsten N, Protz K, Augustin M, Blome C. Social participation of people with chronic wounds: a systematic review. Int Wound J. 2021;18(3):287–311. https://doi.org/10.1111/iwj.13533.
9. 9.Lumley SF, O'Donnell D, Stoesser NE, Matthews PC, Howarth A, Hatch SB, Marsden BD, Cox S, James T, Warren F, Peck LJ, Ritter TG, de Toledo Z, Warren L, Axten D, Cornall RJ, Jones EY, Stuart DI, Screaton G, Eyre DW. Antibody status and incidence of SARS-CoV-2 infection in health care workers. N Engl J Med. 2021;384(6):533–40. https://doi.org/10.1056/NEJMOA2034545.
10. Thoits PA. Perceived social support and the Voluntary, Mixed, or pressured use of mental health services. Soc Mental Health. 2011;1(1):4–19. https://doi.org/10.1177/2156869310392793.
11. Wen B, Musa SN, Onn CC, Ramesh S, Liang L, Wang W, Ma K. The role and contribution of green buildings to sustainable development goals. Build Environ. 2020;185(May):107091. https://doi.org/10.1016/j.buildenv.2020.107091.
12. Al-Sadi AM, Al-Oweisi FA, Edwards SG, Al-Nadabi H, Al-Fahdi AM. Genetic analysis reveals diversity and genetic relationship among trichoderma isolates from potting media, cultivated soil and uncultivated soil. BMC Microbiol. 2015;15(1). https://doi.org/10.1186/s12866-015-0483-8.
13. Agyemang-Duah W. Towards good health and wellbeing: Understanding health conditions and healthcare needs among informal caregivers of older adults in Ghana. J Health Popul Nutr. 2025;44:415. https://doi.org/10.1186/s41043-025-01131-7.
14. Reisinger MW, Moss M, Clark BJ. Is lack of social support associated with a delay in seeking medical care? A cross-sectional study of Minnesota and Tennessee residents using data from the behavioral risk factor surveillance system. BMJ Open. 2018;8:e018139. https://doi.org/10.1136/bmjopen-2017-018139.
15. Park J, Kibe P, Yeboah G, et al. Factors associated with accessing and utilisation of healthcare and provision of health services for residents of slums in low and middle-income countries: a scoping review of recent literature. BMJ Open. 2022;12:e055415. https://doi.org/10.1136/bmjopen-2021-055415.
16. Mensah AA, Amponsah O, Takyi SA, et al. Land governance in Ghana and its implications for inclusive City development. SN Soc Sci. 2022;2:5. https://doi.org/10.1007/s43545-021-00310-7.
17. Gouin JP, Glaser R, Malarkey WB, Beversdorf D, Kiecolt-Glaser J. Chronic stress, daily stressors, and Circulating inflammatory markers. Health Psychol. 2012;31(2):264–8. https://doi.org/10.1037/a0025536.
18. Gyasi RM, Phillips DR, Amoah PA. Multidimensional social support and health services utilization among noninstitutionalized older persons in Ghana. J Aging Health. 2020;32(3–4):227–39. https://doi.org/10.1177/0898264318816217.
19. Gyasi RM, Phillips DR, Abass K. Social support networks and psychological well-being in community-dwelling older Ghanaian cohorts. Int Psychogeriatr. 2019;31(7):1047–57. https://doi.org/10.1017/S1041610218001539.
20. Agyemang-Duah W, Rosenberg MW. Healthcare utilization among informal caregivers of older adults in the Ashanti region of ghana: a study based on the health belief model. Arch Public Health. 2023;81(1):1–18.
21. Agyemang-Duah W, Abdullah A, Rosenberg MW. Caregiver burden and health-related quality of life: A study of informal caregivers of older adults in Ghana. J Health Popul Nutr. 2024;43(1):1–14.
22. Agyemang-Duah W, Rosenberg MW. Geographical location as a determinant of caregiver burden: a rural-urban analysis of the informal caregiving, health, and healthcare survey in Ghana. BMC Prim Care. 2024;25(1):1–17.
23. Agyemang-Duah W, Oduro MS, Peprah P, Adei D, Nkansah JO. The role of gender in health insurance enrollment among geriatric caregivers: results from the 2022 informal caregiving, health, and healthcare survey in Ghana. BMC Public Health. 2024;24:1–14.
24. George ES, Kecmanovic M, Meade T, Kolt GS. Psychological distress among carers and the moderating effects of social support. BMC Psychiatry. 2020;20(1). https://doi.org/10.1186/s12888-020-02571-7.
25. Grunfeld E, Coyle D, Whelan T, Clinch J, Reyno L, Earle CC, Willan A, Viola R, Coristine M, Janz T, Glossop R. Family caregiver burden: results of a longitudinal study of breast cancer patients and their principal caregivers. CMAJ Can Med Association J. 2004;170(12):1795–801. https://doi.org/10.1503/cmaj.1031205.

26. Berglund E, Lytsy P, Westerling R. Living environment, social support, and informal caregiving are associated with healthcare seeking behaviour and adherence to medication treatment: a cross-sectional population study. Health Soc Care Community. 2019;27(5):1260–70. https://doi.org/10.1111/hsc.12758.
27. Lwanga S, Lemeshow S. (1991). Sample size determination in health studies: a practical manual.
28. Agyemang-Duah W, Asante D, Oduro Appiah J, Morgan AK, Mensah IV, Peprah P, Mensah AA. System, institutional, and client-level factors associated with formal healthcare utilisation among older adults with low income under a social protection scheme in Ghana. Arch Public Health. 2023;81(1):68.
29. Agyemang-Duah W, Braimah JA, Asante D, Appiah JO, Peprah P, Awuviry-Newton K, Opoku K. Family support, perceived physical activeness and chronic non-communicable diseases as determinants of formal healthcare utilization among older adults with low income and health insurance subscription in Ghana. Advances in family caregiving policy and research. Routledge; 2025. pp. 127–43.
30. Hailu GN, Gebru HB, Hagos GG, et al. The role of family caregivers in supporting older adults in africa: systematic review. BMC Geriatr. 2025;25:491. https://doi.org/10.1186/s12877-025-06154-7.
31. Kent EE, Mollica MA, Dionne-Odom JN, et al. Effect of instrumental support on distress among family caregivers: findings from a nationally representative study. Palliat Supportive Care. 2020;18(5):519–27. https://doi.org/10.1017/S1478951520000036.
32. Keefe J, Rajnovich B. (2007). To pay or not to pay: examining underlying principles in the debate on financial support for family caregivers. Can J Aging. 26(S1), 77–89
33. Zimet GD, Dahlem NW, Zimet SG, Farley GK. The multidimensional scale of perceived social support. J Pers Assess. 1988;52(1):30–41. https://doi.org/10.1207/s15327752jpa5201_2.
34. Boro B, Saikia N. A qualitative study of the barriers to utilizing healthcare services among the tribal population in Assam. PLoS ONE. 2020;15(10). https://doi.org/10.1371/journal.pone.0240096.
35. Kurichi JE, Pezzin L, Streim JE, Kwong PL, Na L, Bogner HR, Xie D, Hennessy S. Perceived barriers to healthcare and receipt of recommended medical care among elderly medicare beneficiaries. Arch Gerontol Geriatr. 2017;72:45–51. https://doi.org/10.1016/j.archger.2017.05.007.
36. Berkman, Sheridan SL, Donahue KE, Halpern DJ, Crotty K. Low health literacy and health outcomes: an updated systematic review. Ann Intern Med. 2011;155(2):97–107. https://doi.org/10.7326/0003-4819-155-2-201107190-00005.
37. Loiselle CG, Dubois S. The impact of a multimedia informational intervention on healthcare service use among women and men newly diagnosed with cancer. Cancer Nurs. 2009;32(1):37–44. https://doi.org/10.1097/01.NCC.0000343371.24517.66.
38. Salinas-Rodríguez A, Moreno-Tamayo K, Hernández-Serrato M, Enríquez-Rosas M del R., Manrique-Espinoza BS. (2018). Multidimensional social support is associated with healthcare utilization among older Mexican adults. Eur J Ageing, 15(1), 77–85. https://doi.org/10.1007/s10433-017-0428-1
39. Oduro Appiah J, Agyemang-Duah W, Peprah C, Adei D, Peprah P, Fordjour AA. Transportation barriers to formal healthcare utilisation and associated factors among poor older people under a social protection programme in Ghana. J Transp Health. 2020;19:100965.
40. Agyemang-Duah W, Peprah C, Peprah P. Barriers to formal healthcare utilisation among poor older people under the livelihood empowerment against poverty programme in the Atwima Nwabiagya district of Ghana. BMC Public Health. 2019;19(1):1185.
41. Rosenstock IM. Historical origins of the health belief model. Health Educ Monogr. 1974;2(4):328–35.
42. Henshaw EJ, Freedman-Doan CR. Conceptualizing mental health care utilization using the health belief model. Clin Psychol. 2009;16(4):420.


## Publisher's Note